

Infosys Virtual internship

STUDENT'S WEEKLY DIARY

Week-1		DATE	2 Feb 2026
Dept./Division	DATA VISUALISATION	Name of finished Product	YouTube Channel Performance and Engagement Analytics Dashboard
Name of Supervisor With e-mail id	Akhil Mohammed Hussain springboardmentor210@gmail.com		
Main points of the Week			
Studied project requirements and workflow. Set up Python development environment. Created virtual environment and installed required libraries like Streamlit, Pandas, SQLAlchem, PyMySQL, and Google API client. Created project folder structure for data processing, database, and Streamlit app. Initialized Git repository for version control. Created .env file to store API credentials securely. Tested setup using a basic “Hello World” Streamlit app. Designed basic Streamlit homepage and Channel ID input interface.			

Week-2		DATE	9 Feb 2026
Dept./Division	DATA VISUALISATION	Name of finished Product	YouTube Channel Performance and Engagement Analytics Dashboard
Name of Supervisor With e-mail id	Akhil Mohammed Hussain springboardmentor210@gmail.com		
Main points of the Week			
Created YouTube Data API v3 credentials using Google Cloud Console. Developed Python module for YouTube API authentication. Stored API key securely using environment variables. Tested API connection using sample Channel ID. Implemented channel data extraction function. Retrieved channel name, description, subscribers, total views, video count, creation date, and thumbnail. Added error handling for invalid Channel ID and API issues. Converted extracted channel data into Pandas DataFrame.			

Week-3		DATE	16 Feb 2026
Dept./Division	DATA VISUALISATION	Name of finished Product	YouTube Channel Performance and Engagement Analytics Dashboard
Name of Supervisor With e-mail id	Akhil Mohammed Hussain springboardmentor210@gmail.com		
Main points of the Week			
Developed Python functions to insert extracted YouTube data into the SQL database. Stored channel details, video metadata, and video statistics in separate tables. Implemented foreign key relationships to connect channel and video data properly. Added validation to prevent duplicate records and maintain data accuracy. Created logic to check whether a channel already exists before inserting new data. Integrated YouTube API data extraction and database storage with the Streamlit interface. Added progress indicators for fetching channel data, fetching video details, and storing records. Displayed success/error messages and summary of total videos processed and stored. Added feature to view recently analyzed channels from the database.			

Week-4		DATE	23 Feb 2026
Dept./Division	DATA VISUALISATION	Name of finished Product	YouTube Channel Performance and Engagement Analytics Dashboard
Name of Supervisor With e-mail id	Akhil Mohammed Hussain springboardmentor210@gmail.com		
Main points of the day/Week			
Developed SQL analytics queries to generate insights from stored YouTube data. Created queries for top 10 most viewed videos, highest engagement videos, posting frequency, and monthly upload trends. Calculated important metrics such as engagement rate, average views per video, subscriber-view ratio, and content performance score. Converted query results into Pandas DataFrames for analysis and visualization. Designed the main Streamlit dashboard overview page. Added metric cards to show total subscribers, total videos, total views, and average engagement rate. Added filters for channel selection and date range analysis. Created interactive visualizations such as line chart, bar chart, pie chart, scatter plot, heatmap, histogram, and multi-line chart. Added chart labels, legends, and hover details to make the dashboard easy to understand.			

Week-5		DATE	2 Mar 2026
Dept./Division	DATA VISUALISATION	Name of finished Product	YouTube Channel Performance and Engagement Analytics Dashboard
Name of Supervisor With e-mail id	Akhil Mohammed Hussain springboardmentor210@gmail.com		
Main points of the day/Week			
Implemented search and filter functionality in the Streamlit dashboard. Added search bar to find videos by title or description. Created filters for view count, date range, engagement rate, and video duration. Added multi-select filters that can work together. Displayed filtered video results in a sortable table. Added “Clear Filters” button to reset all selected filters. Developed comparative analytics module for comparing multiple channels. Created side-by-side channel comparison for subscribers, views, videos, and engagement rate. Added leaderboard feature to rank channels based on performance metrics. Added export option to download comparison reports in CSV or PDF format.			

Week-6		DATE	9 Mar 2026
Dept./Division	DATA VISUALISATION	Name of finished Product	YouTube Channel Performance and Engagement Analytics Dashboard
Name of Supervisor With e-mail id	Akhil Mohammed Hussain springboardmentor210@gmail.com		
Main points of the day/Week			
Implemented data export and reporting features. Added options to download analytics data in CSV and Excel formats. Created PDF report generation with key metrics, charts, and insights. Added custom report builder to select required metrics and visualizations. Improved application performance using Streamlit caching. Optimized database queries and data loading for large YouTube channels. Added error handling for API calls, database operations, and user inputs. Created user-friendly error messages and validation feedback. Added loading states, progress bars, tooltips, and help section for better user experience. Started testing and documentation for final project submission.			

Week-7		DATE	16 Mar 2026
Dept./Division	DATA VISUALISATION	Name of finished Product	YouTube Channel Performance and Engagement Analytics Dashboard
Name of Supervisor With e-mail id	Akhil Mohammed Hussain springboardmentor210@gmail.com		
Main points of the day/Week			
Performed testing of YouTube API data extraction. Checked accuracy of channel details, video metadata, and engagement metrics. Tested Streamlit dashboard pages and navigation. Verified charts, graphs, filters, search option, and comparison features. Tested SQL database connection and data retrieval. Checked whether duplicate data is avoided during repeated channel analysis. Reviewed error handling for invalid Channel ID, API errors, and empty data. Improved dashboard layout, labels, and user-friendly messages.			

Week-8		DATE	23 Mar 2026
Dept./Division	DATA VISUALISATION	Name of finished Product	YouTube Channel Performance and Engagement Analytics Dashboard
Name of Supervisor With e-mail id	Akhil Mohammed Hussain springboardmentor210@gmail.com		
Main points of the day/Week			
Performed final end-to-end testing of the complete application. Verified complete workflow from Channel ID input to report generation. Reviewed accuracy of calculated metrics like engagement rate and average views. Prepared final project documentation. Documented development process including YouTube API integration, database storage, analytics, dashboard, and export features. Added screenshots, workflow explanation, and module details in the final report. Mentioned application features and future enhancements. Prepared final report for submission.			

Signature of Industry Supervisor